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**Analysing a Student's Use of ChatGPT for
University Studies Through Automated
Conversation Classification**

Bachelor's Thesis (9 ECTS)

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Abstract:

This bachelor thesis analyses one student's use of ChatGPT during computer science studies by using an exported OpenAI conversation history as behavioural trace data. The aim is to describe how ChatGPT was used across academic tasks and to examine whether usage patterns differed between courses. The method follows a scripted pipeline in which conversations are extracted from the OpenAI JSON export, classified by academic relevance, analysed qualitatively with controlled labels, and the resulting tagged relevant data is aggregated quantitatively. The final analysed dataset contains 702 academically relevant or mixed conversations. The results show that the use of ChatGPT was mainly explanation-oriented, while debugging and code generation were also common but secondary. Course-level results show that mathematically oriented courses were dominated by explanations, whereas several programming-oriented courses had relatively larger code-generation or debugging components. Conversation volume by course did not show a meaningful positive relationship with grades. The thesis contributes a detailed single-student case study and a reproducible workflow for analysing exported ChatGPT conversation histories.

Keywords: generative artificial intelligence, ChatGPT, learning analytics, conversation classification, computer science education

CERCS: P170 Computer science, numerical analysis, systems, control

Üliõpilase ChatGPT kasutuse analüüs

ülikooliõpingutes automaatse

vestlusklassifikatsiooni abil

Lühikokkuvõte:

See bakalaureusetöö analüüsib ühe üliõpilase ChatGPT kasutust informaatikaõpingutes, kasutades OpenAI eksporditud vestlusajalugu. Töö eesmärk on kirjeldada, kuidas ChatGPT-d kasutati akadeemiliste ülesannete toetamiseks, ning uurida, kas kasutusmustrid erinesid kursuste lõikes. Meetod põhineb skriptidel: toorandmetest eraldatakse vestlused, vestlused klassifitseeritakse akadeemilise asjakohasuse järgi, asjakohaseid vestlusi analüüsitakse kvalitatiivselt kontrollitud märgenditega ning märgendatud asjakohased andmed koondatakse kvantitatiivseks analüüsiks. Lõplik analüüsitud andmestik sisaldab 702 akadeemiliselt asjakohast vestlust. Tulemused näitavad, et ChatGPT kasutus oli peamiselt selgitustele suunatud, kuigi silumine ja koodi genereerimine olid samuti levinud. Kursuste lõikes olid matemaatilise sisuga kursused valdavalt selgituskesksed, samal ajal kui mitmes programmeerimisega seotud kursuses oli suhteliselt suurem koodi genereerimise või silumise osakaal. Kursusepõhine vestluste arv ei näidanud positiivset seost hinnetega. Töö panus on detailne üksikjuhtumi analüüs ja taaskasutatav töövoog ChatGPT vestluste ekspordi analüüsimiseks.

Võtmesõnad: generatiivne tehisintellekt, ChatGPT, õpianalüütika, vestluste klassifitseerimine, informaatika

CERCS: P170 Arvutiteadus, arvutusmeetodid, süsteemid, juhtimine (automaatjuhtimisteooria)

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1. Introduction

Generative artificial intelligence (GenAI) tools based on large language models (LLMs), such as ChatGPT, have rapidly become part of students' study routines. In higher education, these systems are increasingly used for interactive explanations, planning, and step-by-step support, but they also introduce risks: LLM responses can be misleading or incorrect while sounding confident, and students may develop shallow understanding if they accept outputs uncritically. Because of this dual nature, understanding how students use GenAI, rather than whether they use it, is important for evaluating its role in learning [1].

Empirical research already suggests that there is no single typical pattern of student use. Survey-based work indicates that students rely on ChatGPT in different ways, including profiles characterized by low use, learning-focused use, and more assignment-delegation-oriented use. Such heterogeneity implies that educational implications likely depend on usage strategies rather than the technology alone [2].

In computer science and programming contexts, the question becomes even more concrete: students may use GenAI to diagnose errors, understand code, or solve practical tasks under time pressure. The study by Lepp and Kaimre [3] reports that students commonly use AI assistants for programming-related support and examines how reported use relates to performance, raising the possibility that frequent reliance does not automatically translate to better learning outcomes. This makes course-level context and task type central for analysis in computer science studies.

However, a key limitation of much existing evidence is methodological. Many studies measure student GenAI use primarily through questionnaires or interviews, which are valuable for attitudes and perceptions but can be weak measures of actual behaviour. Educational measurement research has shown that self-reports and log or trace data may diverge, meaning that what learners say they did does not always align with what is visible in behavioural records [4]. This motivates complementary approaches that observe usage directly when possible.

This thesis addresses that gap with an evidence-based, in-depth single-student case study. The dataset consists of the author's complete ChatGPT conversation history over an extended period of university study, obtained from an OpenAI data export in JSON format. Importantly, these conversations were not produced with the intention of creating a thesis dataset at the time of use; therefore, they reflect naturalistic study behaviour rather than interaction shaped by

participation in a study. Digital-trace approaches have long been recognized as a way to observe behaviour at scale and over time, while also raising questions of interpretation and privacy that must be handled carefully [5].

Because a multi-year conversation archive is too large to analyse manually with consistent criteria, this thesis develops an automated conversation analysis pipeline. Conversations are first classified by academic relevance and then analysed qualitatively with structured labels, including usage type and course association, to enable quantitative summaries and targeted inspection. Recent methodological work suggests that LLMs can serve as scalable annotators for text classification tasks and can reduce the cost of producing consistent labels, although careful schema design and critical interpretation of annotation quality remain necessary [6, 7].

The thesis is driven by the following research questions:

- RQ1. How did the author use ChatGPT during computer science studies and across different courses and tasks? This includes frequency of use and the distribution of interaction types across academic tasks.
- RQ2. Did the author use ChatGPT differently in different courses? This includes statistical comparison across courses and investigation of outlier courses through closer inspection to propose plausible explanations grounded in course context and interaction traces.

This work contributes (1) a behavioural, data-driven account of one student's GenAI-supported study practices and (2) a reproducible workflow, including scripts and pipeline structure, that can be adapted for similar conversation-history datasets. The Results chapter reports only findings from this dataset, while the Discussion chapter interprets those findings and compares them with prior survey-based evidence to evaluate whether the case study reconfirms, complements, or contradicts previously reported patterns of student GenAI use [2, 3].

The remainder of the thesis is structured as follows. Chapter 2 reviews related work on student GenAI use in higher education, focusing primarily on questionnaire-based and interview-based studies and motivating the categories used in the analysis. Chapter 3 describes the data source and the conversation classification and analysis pipeline. Chapter 4 presents the results as descriptive statistics and course-level comparisons. Chapter 5 discusses the implications of the findings, limitations of the approach, and threats to validity. Chapter 6 concludes the thesis and outlines possible future work.

The accompanying scripts and configuration files are listed in Appendix A and submitted as supplementary material.

AI-based tools were used during the preparation of this thesis for formal and supporting purposes, including grammar checking, wording refinement, and consistency checking. ChatGPT and Codex were used as supporting tools, while the author reviewed and approved the final text and interpretations.

2. Background

Generative artificial intelligence (GenAI), especially large language models such as ChatGPT, has become increasingly relevant in higher education because such systems can generate explanations, code examples, summaries, and answers to natural-language questions. Kasneci et al. [1] describe large language models as tools with significant educational potential, especially because they can provide immediate feedback, explanations, and personalised support. At the same time, they also emphasize risks such as incorrect outputs, overconfidence, bias, and the possibility that learners may rely on generated answers without sufficient critical evaluation. This dual role makes GenAI especially relevant for computer science education, where students often work through iterative problem-solving tasks and may use AI systems for debugging, code explanation, or conceptual clarification.

Existing research shows that students do not use ChatGPT in only one way. Stojanov et al. [2] examined university students' self-reported reliance on ChatGPT for learning and identified different patterns of reliance. Their results suggest that students' use of ChatGPT varies depending on the type of task and the level of dependence on the tool. This is important for the present thesis because it supports treating GenAI use as a set of different practices rather than as one uniform activity. In other words, it is not sufficient to ask only whether a student used ChatGPT; it is also necessary to ask what the tool was used for and how that use varied across learning situations.

The study by Lepp and Kaimre [3] is the most directly relevant point of comparison for this thesis, as it examines students' perceptions, reported use, and performance in an Object-Oriented Programming course. Their study found that students primarily used AI chatbots for programming-related tasks. They also reported that the frequency of chatbot use was negatively correlated with course performance. This does not necessarily mean that AI use directly caused

weaker performance, but it does suggest that heavier reported use is not automatically associated with better outcomes. The study by Lepp and Kaimre [3] is especially relevant because it focuses on programming education at the University of Tartu and provides a close point of comparison for interpreting the present thesis.

A limitation of much previous work is that it often relies on self-reported data. Surveys and interviews are valuable for studying students' perceptions, attitudes, and reported habits, but they may not accurately reflect actual behaviour. Han and Yang [4] show that self-reported data and log data may diverge when analysing learning experiences. This distinction is central to the motivation of the present thesis. Instead of asking the author to retrospectively describe how ChatGPT was used, this thesis treats the exported conversation history as a form of behavioural trace data. This makes it possible to examine what was recorded in the ChatGPT interactions, how conversations were distributed across time, and how usage differed between courses.

The use of behavioural trace data also connects the thesis to a broader tradition of computational social science. Lazer et al. [5] argue that digital traces can make it possible to observe behaviour at scale and over time, although such data also require careful interpretation. In the context of this thesis, the OpenAI export is treated as a personal behavioural trace dataset: it contains real interactions that were not created for research purposes at the time of use. Because the conversations were not originally created for this thesis, the dataset reflects ordinary recorded use rather than researcher-designed tasks. The data show interaction with ChatGPT, but not the full study process, and therefore require careful interpretation.

Because the dataset is too large to classify manually in a consistent way, the thesis uses LLM-based annotation as part of the analysis pipeline. Gilardi et al. [6] show that ChatGPT can perform competitively on text annotation tasks, while Tan et al. [7] provide a broader overview of the use of large language models for data annotation and synthesis. These works support the methodological plausibility of using LLMs as scalable annotation tools, provided that the task is clearly defined and the outputs are interpreted critically. In the present thesis, this is implemented through fixed prompts, structured JSON output schemas, and predefined categories for relevance, usage type, sentiment, tags, and course matching.

The present thesis therefore builds on two main strands of previous work. First, it relates to studies of students' GenAI use in learning, especially Stojanov et al. [2] and Lepp and Kaimre [3]. Second, it relates to methodological work on behavioural trace data and automated

annotation, especially Han and Yang [4], Lazer et al. [5], Gilardi et al. [6], and Tan et al. [7]. The contribution of this thesis is to combine these perspectives in a single-student case study: the work analyses actual ChatGPT conversation history rather than self-reported use, and it uses automated classification to turn a large personal archive into structured data for analysing study-related AI use.

3. Method

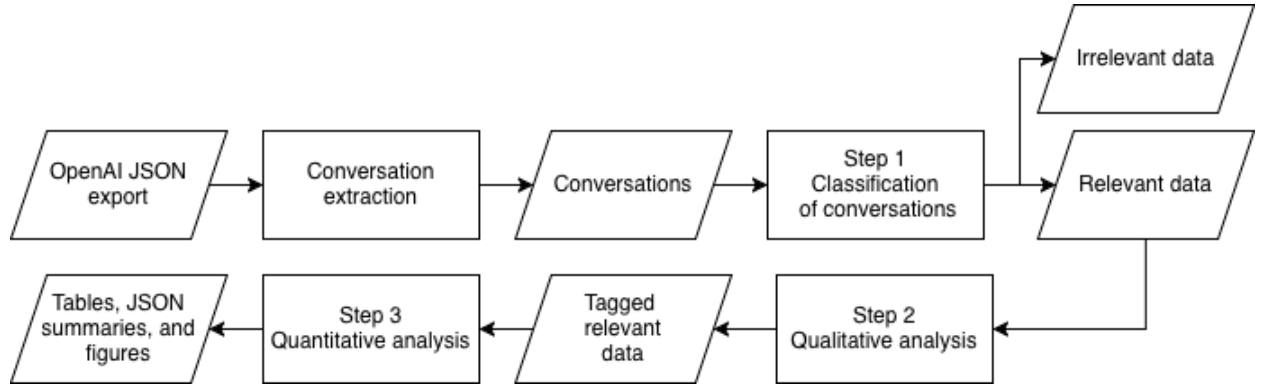


Figure 1. Overview of the conversation classification and analysis workflow.

This chapter describes the method used to answer RQ1 and RQ2. Figure 1 summarizes the full pipeline followed in the thesis: OpenAI JSON export -> conversation extraction -> conversation-level data -> classification of conversations -> relevant data -> qualitative analysis -> tagged relevant data -> quantitative analysis -> tables, JSON summaries, and figures. The pipeline is organised so that each stage has a defined input, process, and output.

The method has three main analysis steps after data extraction. Step 1 is classification of conversations by academic relevance. Step 2 is qualitative analysis of the relevant conversations with controlled labels. Step 3 is quantitative analysis of the tagged relevant data. The outputs from Steps 1 and 2 define the dataset used to answer RQ1, while the quantitative outputs from Step 3 are used to answer both RQ1 and RQ2.

3.1. Data Source and Unit of Analysis

The raw data is obtained by submitting a data access request to OpenAI and downloading the resulting export archive. The export contains account-level activity, including conversations and other assets. For this thesis, the focus is exclusively on textual conversation history.

The conversation data is stored in JSON format in the export package. The pipeline expects a file containing a list of conversation objects, or a top-level wrapper object containing such a list. The unit of analysis is a single ChatGPT conversation as represented in the export JSON. Each conversation object contains metadata, timestamps, and message structure. The pipeline performs classification at the conversation level, not at the individual-message level.

The export contains content authored by multiple roles, including user and assistant messages, and may include system-level or platform metadata. During classification, the full conversation object is provided to the classifier model as JSON, while the classifier prompt instructs the model to focus on textual content and ignore irrelevant metadata fields. During qualitative analysis, the pipeline constructs the input as concatenated user-authored prompts, and sentiment is defined as sentiment of the user prompts rather than the assistant responses.

3.2. Data Extraction

Input: the raw OpenAI data export in JSON format.

Process: conversations are extracted from the export structure and represented as conversation-level objects. The extraction keeps the conversation text and metadata needed for later classification, while other export assets are outside the scope of this thesis.

Output: extracted conversation objects that can be passed to the classification step. This output is the first structured data object in the pipeline shown in Figure 1.

3.3. Step 1: Classification of Conversations

Input: extracted conversation objects from the data extraction step.

Process: classification of conversations is implemented as a structured prompting task. Each conversation is sent independently to an LLM with a fixed instruction prompt that requires a strict JSON label object. Classification uses OpenAI's Responses API from the official Python client. The model identifier in the implemented scripts is `gpt-5-mini`, and the same model, prompt, and output schema are used across the dataset. No additional sampling parameters, such as temperature or top_p, are explicitly specified, so the API defaults are used.

The classification prompt defines the main relevance labels `relevant_to_university_studies`, `personal_irrelevant`, and `mixed`. The dataset is not filtered manually by selecting study conversations. Instead, filtering is performed algorithmically via this classification step. The classification output also contains auxiliary labels such as `domain`, `broad_usage_types`, `discard_reason`, and `notes`, but the relevance label is the main basis for dataset inclusion.

Output: a classified conversation dataset in which each conversation has a relevance label and auxiliary classification fields. Conversations labelled `personal_irrelevant` are stored separately

and are not used for academic analysis. Conversations labelled `relevant_to_university_studies` and `mixed` are passed to qualitative analysis.

3.4. Step 2: Qualitative Analysis of Relevant Conversations

Input: conversations classified as academically relevant or mixed after Step 1.

Process: qualitative analysis adds variables that describe how the author used ChatGPT in academic settings. The prompt uses concatenated user-authored messages and a candidate course list for the relevant semester window. The system prompt enforces a strict JSON schema and a controlled vocabulary for the labels. The category structure was informed by supervisor guidance and ongoing project work, but the thesis does not cite unpublished work for these categories.

The qualitative analysis assigns sentiment of user prompts, controlled usage tags, and course matches. Sentiment is coded as positive, neutral, negative, or mixed. Usage tags are assigned from the controlled set code generation, debugging, and explanations. Course matching connects conversations to course instances described by course code, course name, semester dates, and course description. The model selects the single best matching course where possible, or multiple courses only if the conversation clearly spans more than one subject.

Output: tagged relevant data. This output contains the relevant conversations with sentiment labels, usage tags, and matched course metadata. It provides the qualitative basis for describing what kinds of academic support ChatGPT was used for.

3.5. Tagged Relevant Data

The data object between qualitative analysis and quantitative analysis is the tagged relevant data. It contains only the academically relevant or mixed conversations selected after classification, together with the qualitative labels assigned in Step 2. This object is important because it defines the final dataset from which the results are computed. It is also the point at which the raw conversation archive has been transformed into structured data suitable for aggregation.

3.6. Step 3: Quantitative Analysis and Outputs

Input: tagged relevant data from Step 2.

Process: quantitative analysis aggregates conversation-level labels into descriptive statistics and course-level summaries. For RQ1, the pipeline computes relevance frequencies, usage-type frequencies, tag frequencies, sentiment distribution, temporal distribution of conversation counts, and prompt-level statistics such as the number of user prompts and total prompt length per conversation. Prompt statistics are derived from user-authored prompts only.

For RQ2, the pipeline aggregates the same tagged relevant data at course level. These aggregations include conversation counts by course, tag usage by course, usage pattern summaries by course, and correlations between conversation volume and course outcomes. Course-level pattern comparison focuses not only on raw volume but also on the relative balance between explanations, code generation, and debugging.

Output: JSON and CSV tables and visual figures used in Chapter 4. The outputs include overall counts, tag and use-type summaries, monthly conversation counts, conversation-length summaries, course-level conversation counts, normalized course-tag summaries, and grade-usage correlation outputs.

3.7. Use of Outputs for RQ1 and RQ2

RQ1 asks how the author used ChatGPT during computer science studies and across different courses and tasks. The output from Step 1 identifies which conversations belong to the academic dataset. The output from Step 2 describes those relevant conversations using usage types, sentiment, usage tags, and course matches. The Step 3 quantitative outputs summarize these labels as frequencies, distributions, and figures. Together, these outputs answer RQ1 by showing what kinds of academic tasks and interaction types dominate the dataset.

RQ2 asks whether the author used ChatGPT differently in different courses. The output from Step 2 supplies the matched course metadata and tag labels needed for course comparison. The Step 3 course-level outputs compare conversation volume and normalized tag distribution between courses. These outputs answer RQ2 by showing whether differences are only differences in the amount of use or also differences in the type of use.

3.8. Reliability, Quality Control, and Limitations

The core internal consistency mechanism is that every conversation is processed using the same model, the same system prompts and output schemas, and the same JSON-only output constraint. This reduces within-run inconsistency caused by changing rubrics.

A small number of conversations were manually reviewed to confirm that classifier outputs aligned with human judgement at a coarse level. This served as a sanity check before running the full dataset. No formal inter-rater reliability study was conducted, and the workflow should therefore be understood as an automated classification pipeline with limited manual auditing rather than a fully validated coding scheme.

Because the workflow relies on an LLM classifier, results may be sensitive to model updates, default sampling parameters, and subtle prompt changes. Prompt wording is fixed and programmatic, and the model choice is fixed in code, which supports reproducibility within the current project environment. However, exact replication at a later date may require pinning additional parameters and documenting model versions more precisely.

LLM-based classification should also be treated as a form of automated interpretation. Although the schema enforces structure, category assignment still reflects the model's inference over text. This is mitigated through transparent schemas, mixed-relevance labels, multi-label usage types, and preserved notes for ambiguous cases.

3.9. Ethical and Privacy Considerations

The dataset contains personal interaction history and may include sensitive information. The analysis is conducted on the author's own data. Outputs are aggregated for reporting, and methodological reporting focuses on usage patterns rather than identifiable personal content. Any example conversations included in the thesis are paraphrased or anonymised to avoid unintended disclosure of personal or sensitive information.

4. Results

This chapter presents the findings in direct relation to the two research questions. Section 4.1 addresses RQ1 by describing the overall ways in which the author used ChatGPT during computer science studies. Section 4.2 addresses RQ2 by examining whether that usage differed across courses. The chapter reports only findings from the author's dataset; comparisons with earlier studies are reserved for the Discussion chapter.

4.1. Results for RQ1: How Did the Author Use ChatGPT During Computer Science Studies and Across Different Courses and Tasks?

4.1.1. Dataset Overview and Labelling Outcomes

The results reported here are based on the academically relevant subset produced by the pipeline. The final analysed dataset contains 702 conversations, of which 690 (98.3%) were labelled `relevant_to_university_studies` and 12 (1.7%) as `mixed`. Course matching was successful for most conversations; 45 conversations (6.4%) remained unmatched to a specific course instance.

In terms of annotated academic domain, conversations were most frequently classified as programming (477), followed by math (231) and other (53). Because domain is multi-label at the conversation level, these counts are not mutually exclusive.

4.1.2. Dominant Interaction Types: Tags and Usage Types

The qualitative analysis assigns the controlled tags explanations, debugging, and code generation. Across conversations, the most frequent tag is explanations (648 conversations), followed by debugging (229) and code generation (217). Because tags are multi-label, a single conversation may contribute to multiple counts.

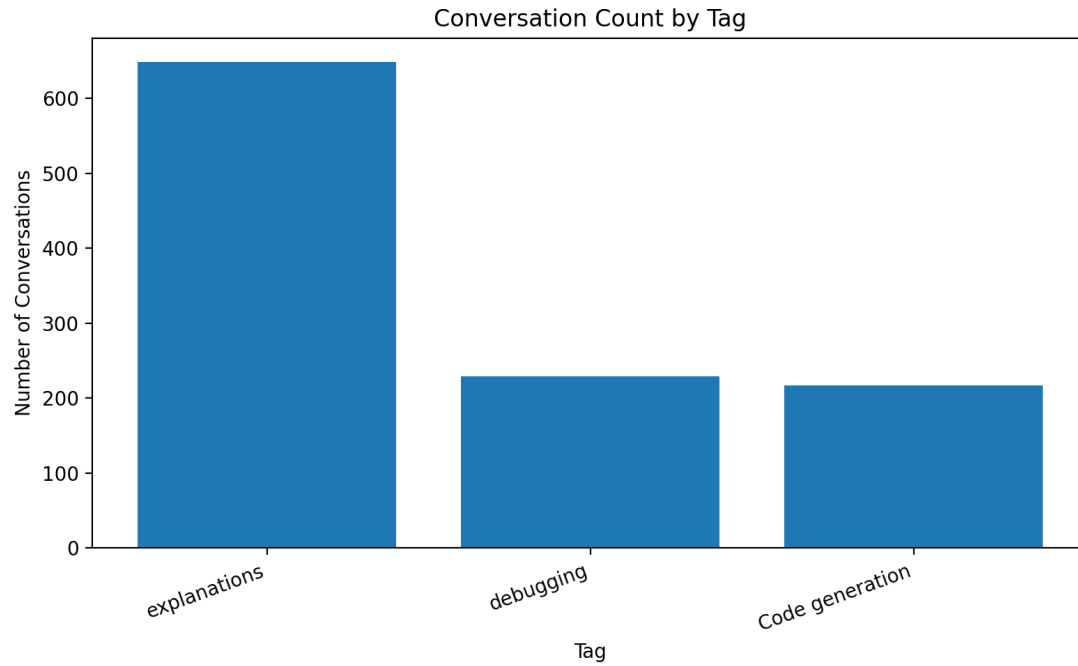


Figure 2. Conversation count by tag.

Figure 2 shows a clear overall pattern: within the academic subset, explanation-seeking is the dominant interaction type, while debugging and code generation are substantial but secondary.

The broader use-type annotation shows the same tendency. The most frequent use types in the dataset are:

- solving_homework_tasks: 546
- understanding_code_examples: 421
- other_academic_use: 222
- studying_for_test: 139
- answering_lecture_quizzes: 63
- non_academic_use: 5

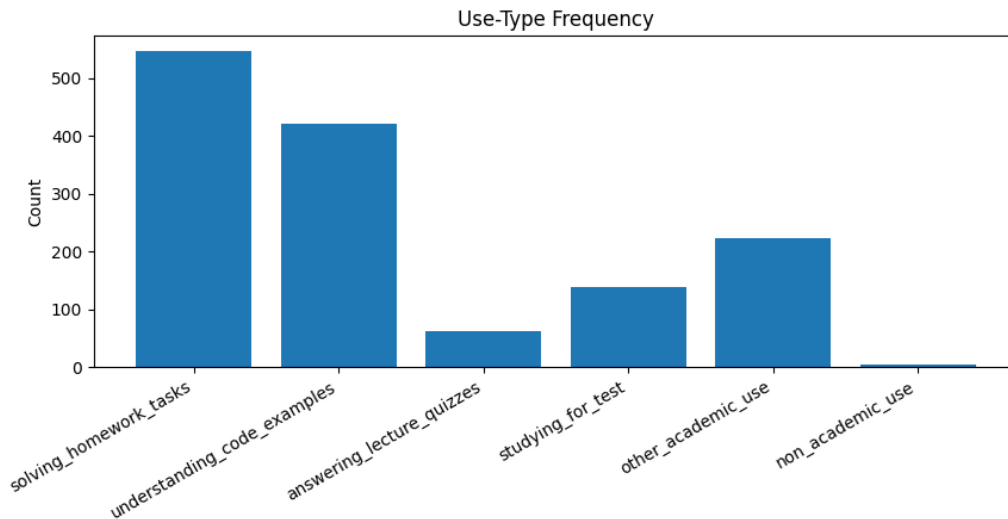


Figure 3. Use-type frequency.

Figure 3 shows that assignment-oriented support and understanding technical material are the dominant broader use types, while exam preparation and lecture-quiz support are present but less central. Because use types are multi-label, totals exceed the number of conversations.

4.1.3. Sentiment of User Prompts

Sentiment was labelled based on user-authored prompts only. The distribution was:

- neutral: 631
- negative: 55
- mixed: 10
- positive: 6

The overwhelming majority of prompts are therefore neutral and task-oriented. Negative sentiment forms a minority and likely reflects moments of friction such as confusion or debugging difficulties.

4.1.4. Temporal Dynamics and Conversation Length

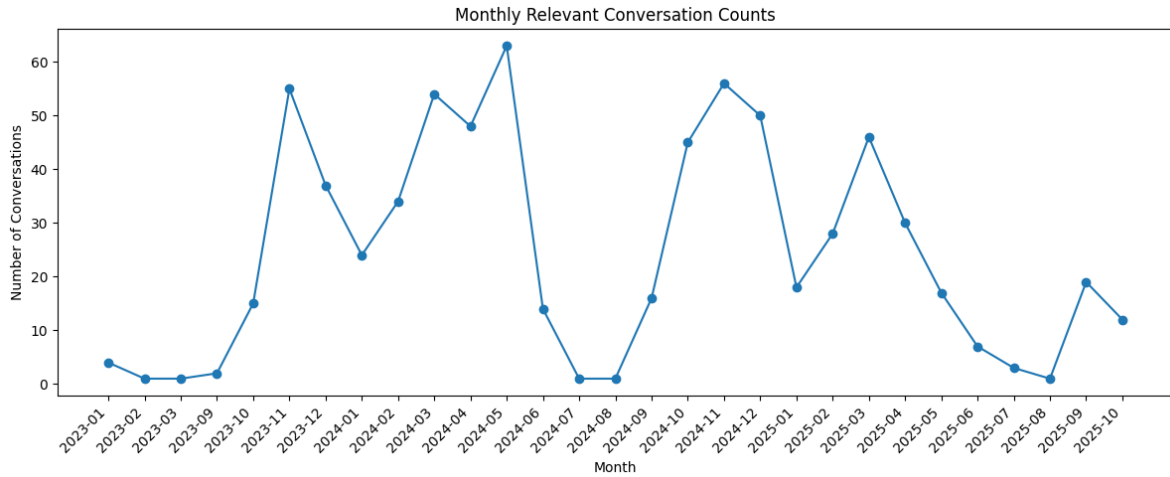


Figure 4. Monthly conversation counts.

Figure 4 shows that relevant usage was not evenly distributed over time. There is a relatively quiet period early in the study history, followed by a clear increase from late 2023 onward and several later peaks in activity. These peaks correlate with semesters, as usage falls during breaks. This suggests that ChatGPT use became more integrated into study practice over time, but also that usage was episodic rather than constant.

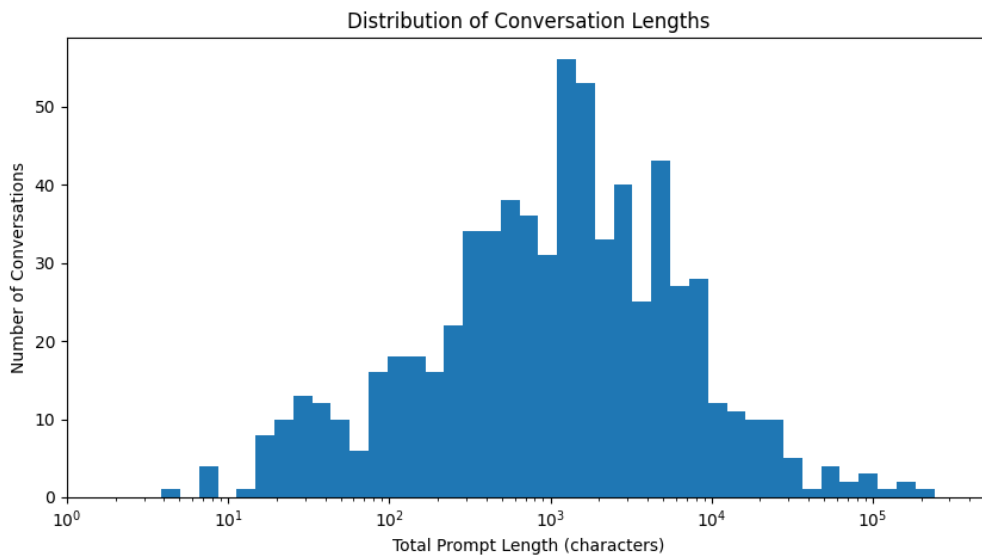


Figure 5. Distribution of conversation lengths measured by total user prompt length.

Figure 5 shows that conversation length, measured as total user prompt length in characters, is strongly skewed. Most interactions are short, but a smaller number of conversations are very large. This suggests that the dataset contains both quick, task-focused prompts and more extended multi-step interactions such as iterative debugging or detailed problem solving.

4.1.5. Answer to RQ1

Based on these results, RQ1 can be answered as follows: the author used ChatGPT primarily as an academic support tool for coursework in computer science and mathematics. The most common forms of use were solving homework-related tasks and understanding code or concepts, while the dominant tag pattern shows that the assistant was used most often for explanations rather than for direct code generation alone. Interaction tone was overwhelmingly neutral and task-focused, suggesting pragmatic use rather than highly emotional engagement. Usage also varied across time, with clear peaks during certain study periods, and most conversations were short even though a smaller number of long, complex interactions also occurred.

4.2. Results for RQ2: Did the Author Use ChatGPT Differently in Different Courses?

4.2.1. Differences in Conversation Volume Across Courses

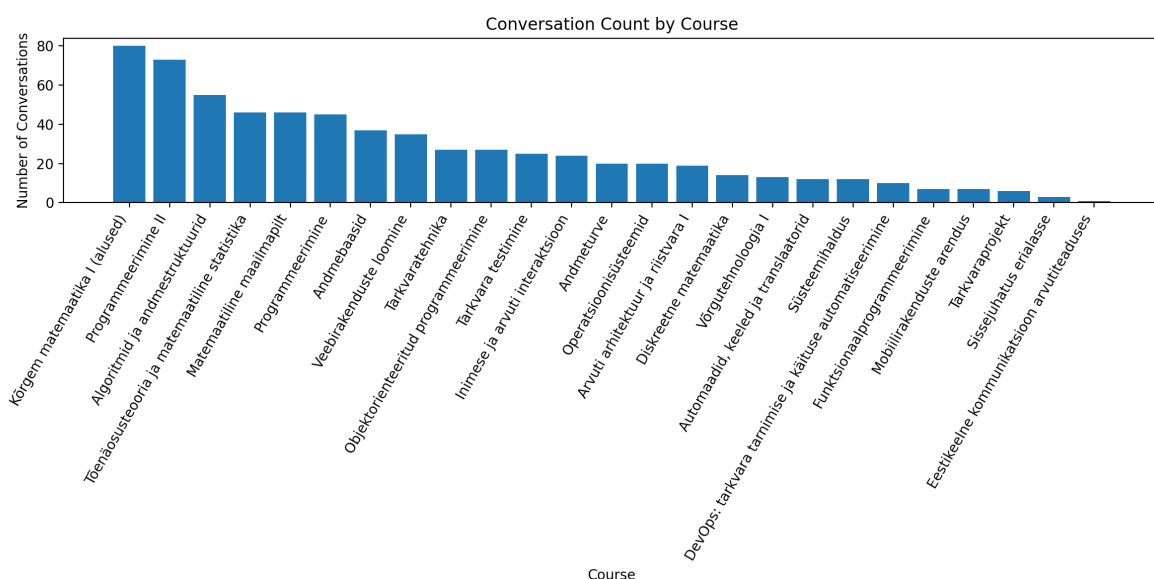


Figure 6. Conversation count by course.

Figure 6 shows that course-matched usage is concentrated in a relatively small number of courses. The largest conversation counts were observed in:

- Higher Mathematics I (Foundations): 80 conversations (11.4%)
- Programming II: 73 (10.4%)
- Algorithms and Data Structures: 55 (7.8%)

- Probability Theory and Mathematical Statistics: 46 (approximately 6.6%)
- Mathematical Worldview: 46 (approximately 6.6%)
- Programming I: 45 (approximately 6.4%)

Beyond these, a long tail of courses accounts for smaller shares. This shows that ChatGPT use was not evenly distributed across the curriculum. Instead, a limited set of core courses, especially mathematics- and programming-heavy ones, accounted for a large share of the activity.

4.2.2. Differences in Course-Level Usage Patterns

Course-level pattern summaries show that the same overall tool was not used in exactly the same way across courses. Although explanations dominate the dataset overall, the balance between explanations, debugging, and code generation varies by course.

Relative Distribution of Conversation Tags in the Top 15 Courses

Values are normalized within each course across explanations, code generation, and debugging.

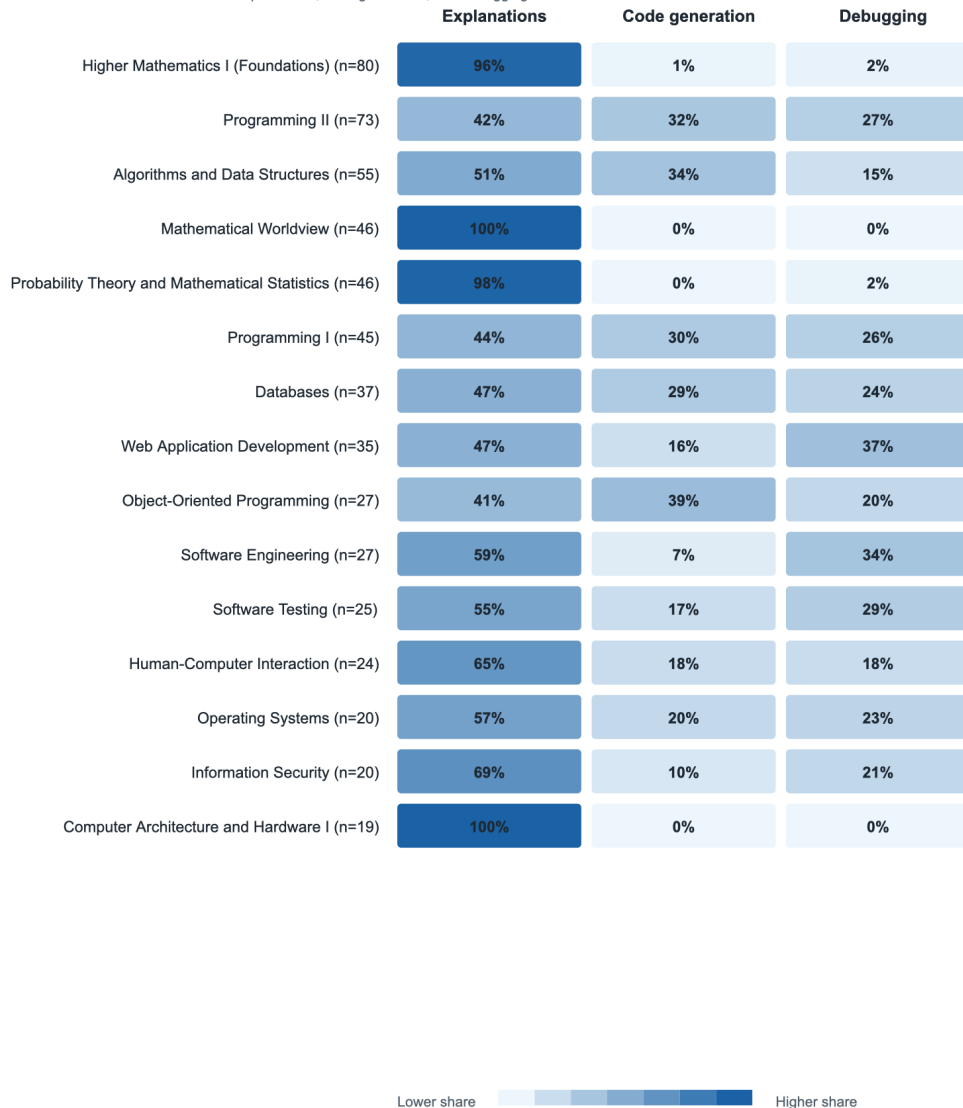


Figure 7. Normalised distribution of explanations, code generation, and debugging in the top 15 courses by conversation count. Values are normalised within each course across the three tags.

Figure 7 uses a normalized heatmap because RQ2 concerns differences in usage pattern, not only raw volume. Each row is normalized within the course across explanations, code generation, and debugging. Therefore, the figure shows the relative distribution of tag types within each course rather than the absolute number of conversations.

Higher Mathematics I (Foundations), Mathematical Worldview, Probability Theory and Mathematical Statistics, and Computer Architecture and Hardware I are strongly explanation-oriented. In these courses, conversations are dominated by explanation-seeking, with little or no code generation and minimal debugging. This is consistent with their more theoretical or conceptual character.

In contrast, several programming-oriented courses show heavier code-generation or debugging components. Programming II, Algorithms and Data Structures, Programming I, Databases, and Object-Oriented Programming have relatively larger code-generation shares, while Web Application Development, Software Engineering, Software Testing, Operating Systems, and Information Security show relatively stronger debugging patterns. This suggests that the functional role of ChatGPT changed with course context: in some courses it was used more as a conceptual explainer, whereas in others it was used more as a practical coding or debugging assistant.

The course-pattern outlier analysis also indicates that some courses deviate more noticeably from the overall course-level pattern than others. In other words, not all differences are explained by volume alone; certain courses are distinctive because their type of AI use differs from the dominant profile.

4.2.3. Relationship Between ChatGPT Usage and Course Outcomes

To examine whether heavier ChatGPT usage is associated with higher course grades, the correlation between conversation count per course attempt and grade was computed using a numeric grade mapping. The reported Pearson correlation was $r = -0.091$.

An additional rank-based check produced a similarly weak relationship, Spearman rho approximately -0.086 . In this dataset, conversation volume by course attempt therefore does not show a meaningful positive association with grades.

Concrete examples illustrate why a strong positive relationship is not evident. High usage can coincide with low grades, as in Higher Mathematics I (Foundations), while high grades can also occur with only moderate usage, as in several programming courses. This suggests that simple usage intensity, measured only as number of conversations mapped to a course attempt, is too coarse to explain academic outcomes in a meaningful way.

4.2.4. Answer to RQ2

Based on these results, RQ2 can be answered as follows: yes, the author used ChatGPT differently across courses. The clearest difference is first visible in conversation volume: a small number of courses account for a disproportionate share of the interactions. However, the differences are not only quantitative. Course-level tag and pattern summaries indicate that mathematically oriented courses were more explanation-driven, whereas several

programming-oriented courses were relatively more associated with code generation or debugging. At the same time, conversation volume alone was not meaningfully related to grades. This suggests that course context influenced how ChatGPT was used, not just how often it was used.

5. Discussion

5.1. Comparison With Previous Research

The findings of this thesis are most directly comparable with the study by Lepp and Kaimre [3], because it also focuses on the use of generative AI in programming education. Lepp and Kaimre [3] found that students mainly used AI chatbots for programming-related tasks, and the present thesis broadly corroborates that observation. In this thesis, the most frequent use types were solving homework tasks and understanding code examples, while the most frequent interaction tag was explanations. Debugging and code generation were also common, but they were less frequent than explanation-seeking. This supports the view that GenAI use in computer science studies is not limited to direct code generation, but also includes clarification, task support, and understanding technical material.

The results also align with Lepp and Kaimre [3] in relation to academic performance. In their study, more frequent reported use of AI chatbots was negatively correlated with course performance. In the present thesis, the relationship between conversation volume and grades was also very weak and slightly negative. This similarity should not be interpreted causally, because neither study proves that AI use directly causes weaker outcomes. However, both results argue against a simple assumption that more frequent AI use automatically leads to better academic performance. Instead, the educational value of GenAI appears to depend more on the type and purpose of use than on usage frequency alone.

The present thesis also complements the study by Stojanov et al. [2], who studied self-reported reliance on ChatGPT for learning and identified different profiles of student reliance. Their findings show that students' use of ChatGPT is heterogeneous rather than uniform. The present thesis supports this broader idea from a different methodological angle. Instead of grouping students into reliance profiles, it shows variation within one student's own usage across courses and task types. Mathematically oriented courses were more explanation-driven, while several programming-oriented courses were relatively more associated with debugging and code generation. This suggests that even within a single student's study history, GenAI use can change depending on course context.

The methodological contribution of the present thesis is also connected to Han and Yang [4]. Han and Yang [4] show that self-reported data and log data may diverge when studying

learning experiences. This matters because much research on student GenAI use relies on questionnaires or interviews. Such methods are useful for understanding perceptions, but they do not directly show what students actually asked, when they asked it, or how interactions developed over time. By analysing exported ChatGPT conversations, the present thesis complements survey-based studies with behavioural trace evidence. This makes it possible to observe temporal patterns, course-level differences, conversation length distributions, and tag-based interaction patterns directly from the recorded data.

The present work is also consistent with the broader argument by Lazer et al. [5] that digital trace data can provide new ways of observing behaviour over time. In this thesis, the OpenAI export is used as a personal trace dataset. This makes it possible to analyse naturalistic interactions that were not produced for a study at the time. Compared with retrospective self-report, this gives a more detailed view of recorded ChatGPT interaction behaviour. However, as Lazer et al. [5] also imply for digital trace research more generally, such data must be interpreted carefully. The exported conversations show one channel of activity, but not the entire learning process.

Finally, the use of LLM-based annotation in this thesis is supported by methodological work on language models as annotation tools. Gilardi et al. [6] show that ChatGPT can perform well on text annotation tasks, and Tan et al. [7] describe large language models as useful tools for data annotation and synthesis. The present thesis applies this idea to a personal conversation archive. However, the results should still be understood as structured interpretations rather than objective ground truth. The annotation pipeline made the analysis scalable and consistent, but it did not remove the need for critical interpretation of categories and outputs.

Overall, the present thesis does not simply repeat previous findings, nor does it clearly contradict them. Rather, it corroborates several observations from Lepp and Kaimre [3] and Stojanov et al. [2], while complementing them through a different kind of data. The main added value is granularity: the thesis shows how one student's GenAI use varied across time, courses, and task types using actual conversation records instead of only self-reported descriptions.

5.2. Interpretation of the Findings

Within the academically relevant dataset, the most important overall finding is that ChatGPT use was strongly explanation-oriented. Explanations were the most frequent tag by a clear margin, while debugging and code generation were also common but secondary. The use-type

results support the same interpretation, since solving homework tasks and understanding code examples were the most frequent categories. This suggests that ChatGPT was often used as a tool for clarifying concepts, understanding tasks, and supporting problem solving, rather than only as a direct solution generator.

The course-level results show that ChatGPT use differed not only in amount but also in character. Some courses had many more conversations than others, but the more important finding is that different courses were associated with different interaction patterns. Mathematically oriented courses were more explanation-driven, while several programming-oriented courses had relatively stronger debugging or code-generation components. This indicates that using ChatGPT for studies is not one stable behaviour. Instead, the role of the tool appears to shift depending on course demands and task type.

The weak relationship between conversation volume and grades is also important. In this thesis, heavier use did not correspond to better course outcomes. However, this does not mean that ChatGPT use was harmful. Conversation volume is a coarse measure. A high number of conversations may reflect greater course difficulty, weaker prior knowledge, more active help-seeking, or more complex assignments. Conversely, low use does not necessarily mean that the student understood the material better; it may simply mean that help was obtained elsewhere or was not needed in that course. Therefore, the results suggest that usage quantity alone is not a good indicator of learning quality.

The temporal and conversation-length results support the interpretation that AI use was episodic and task-driven. Usage increased from late 2023 onward and varied across months, suggesting that ChatGPT became more integrated into study practice over time but was not used at a constant level. Conversation length was also strongly skewed: most interactions were short, while a smaller number were much longer. This likely reflects different types of study situations, ranging from quick clarification requests to extended problem-solving or debugging sessions.

The sentiment results further support the view that the interactions were mostly pragmatic. The large majority of conversations were labelled neutral, while negative, mixed, and positive sentiment were much less common. This suggests that the author's prompts were usually task-focused rather than emotionally expressive. In the context of this thesis, sentiment should not be interpreted as a psychological measure, but as a coarse indicator of interaction tone.

Taken together, the results suggest that the most meaningful distinction is not simply whether GenAI was used, or how often it was used, but how it was used in specific academic contexts. Explanation-seeking, debugging, code generation, course context, and interaction length each capture different aspects of use. This supports the broader argument that student GenAI use should be analysed as a set of context-dependent practices rather than as a single behaviour.

5.3. Limitations and Threats to Validity

The most important limitation of this thesis is its scope. The study is based on one student's ChatGPT history. Therefore, the findings cannot be generalized statistically to all students, all computer science students, or all university contexts. The value of the thesis lies in analytical depth rather than representativeness. It shows how one real conversation archive can be analysed and what kinds of patterns such an analysis can reveal.

There are also internal validity limitations. First, the dataset captures only ChatGPT conversations. It does not show the author's full study process, including lecture materials, textbooks, classmates, teaching assistants, web searches, or other tools. Therefore, the analysis should not be interpreted as a complete description of studying. It is a description of one support-seeking channel within the broader study process.

Second, the analysis depends on automated LLM-based annotation. Although the same model, prompts, and schemas were used consistently, labels such as domain, usage type, sentiment, tags, and course matches remain interpretive. Some conversations may have been ambiguous, and some course matches may be uncertain if the text did not clearly indicate the course context. The structured pipeline increases consistency, but it does not make the labels objectively true.

Third, the relationship between the research questions and the available data is imperfect. The research questions ask how GenAI was used and whether use differed across courses. The exported conversations provide strong evidence for observed ChatGPT interaction patterns, but they do not directly reveal the user's motivation, level of understanding, or learning outcome after each interaction. As a result, the thesis can describe usage patterns more confidently than it can explain their causes.

External validity is limited by the single-student design. The author is a computer science student, and the course distribution, study habits, and willingness to use ChatGPT may differ from those of other students. For example, another student might use AI more often for writing,

translation, summarisation, or exam preparation rather than programming and mathematics. Therefore, the findings should be treated as a detailed case study rather than a general model of student GenAI use.

Reliability is also affected by the technical environment. The annotation results may depend on the model version, prompt wording, and API defaults used at the time of processing. The workflow is scripted and reproducible in principle, but exact reproduction later may require the same model behaviour and configuration. Since the model is external and may change over time, perfect reproducibility cannot be guaranteed.

The grade analysis has further limitations. The correlation between conversation volume and grades is exploratory and should not be interpreted causally. More conversations may indicate difficulty, confusion, effort, or greater help-seeking, not necessarily ineffective use. In addition, conversation count does not distinguish between short and long interactions or between different levels of cognitive engagement. Therefore, grade-related results should be interpreted cautiously.

Finally, because the dataset contains personal interaction history, privacy and ethical considerations are important. The analysis focuses on aggregate patterns rather than identifiable conversation content. All examples included in the thesis are paraphrased or anonymised to reduce the risk of disclosing personal or sensitive information.

Overall, these limitations define the scope of the thesis. The study does not claim to provide a general theory of student GenAI use. Instead, it illustrates how exported conversation histories can be analysed systematically and can provide detailed behavioural evidence that complements previous self-report-based research.

6. Conclusion

This thesis analysed one student's use of ChatGPT during computer science studies using an exported OpenAI conversation history. The aim was to understand how GenAI was used across academic tasks and whether usage differed between courses. To do this, the thesis developed an automated analysis pipeline that classifies conversations by academic relevance, applies qualitative analysis to relevant conversations with structured labels, matches conversations to courses, and produces descriptive summaries and course-level comparisons.

The results show that the author's academically relevant ChatGPT use was concentrated mainly in programming and mathematics contexts. The most common use types were solving homework tasks and understanding code examples, while the most frequent interaction tag was explanations. Debugging and code generation were also important, but they were less frequent than explanation-seeking. This suggests that in this dataset ChatGPT was used primarily for clarification, understanding, and problem-solving support rather than only for direct code generation.

The results also show that ChatGPT use differed across courses. Some courses generated substantially more conversations than others, and the type of use also varied. Mathematically oriented courses were more explanation-driven, while several programming-oriented courses were relatively more associated with debugging or code generation. However, conversation volume did not show a meaningful positive relationship with grades. This suggests that the amount of AI use alone is not enough to evaluate its educational value.

The main contribution of the thesis is twofold. First, it provides a detailed behavioural case study of one student's GenAI-supported study practices. Second, it presents a reproducible workflow for analysing exported ChatGPT conversation histories through classification, qualitative analysis, and quantitative summarisation. Compared with the survey- and interview-based studies discussed in this thesis, this approach gives a more direct view of recorded ChatGPT interaction behaviour. While similar approaches may exist or emerge in related work, the present thesis demonstrates that personal conversation exports can be used systematically to study real academic GenAI use in considerable detail.

Future work could apply the same pipeline to multiple students or to a larger dataset in order to examine whether the patterns observed in this case also appear more broadly. The annotation process could also be improved through formal validation, such as human coding of a sample

of conversations or comparison between multiple annotators. Finally, future studies could investigate not only how often students use GenAI, but also how different forms of use relate to learning strategies, self-regulation, and course design.

References

- [1] Kasneci, E., Sessler, K., Küchemann, S., Bannert, M., Dementieva, D., Fischer, F., Gasser, U., Groh, G., Günnemann, S., Hüllermeier, E., Krusche, S., Kutyniok, G., Michaeli, T., Nerdel, C., Pfeffer, J., Poquet, O., Sailer, M., Schmidt, A., Severin, T., Staab, S., Stammer, W., Stürmer, K., van der Smagt, P., & Kasneci, G. (2023). ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and Individual Differences*, 103, 102274. doi:10.1016/j.lindif.2023.102274.
- [2] Stojanov, A., Liu, Q., & Koh, J. H. L. (2024). University students' self-reported reliance on ChatGPT for learning: A latent profile analysis. *Computers and Education: Artificial Intelligence*, 6, 100243. doi:10.1016/j.caeai.2024.100243.
- [3] Lepp, M., & Kaimre, J. (2025). Does Generative AI Help in Learning Programming: Students' Perceptions, Reported Use and Relation to Performance. *Computers in Human Behavior Reports*, 18, 100642. doi:10.1016/j.chbr.2025.100642.
- [4] Han, F., & Yang, J. (2025). Consistency between self-reported and log data to understand students' experience of learning in flipped classrooms. *Scientific Reports*, 15, 34369. doi:10.1038/s41598-025-17052-8.
- [5] Lazer, D., Pentland, A., Adamic, L., Aral, S., Barabási, A.-L., Brewer, D., Christakis, N., Contractor, N., Fowler, J., Gutmann, M., Jebara, T., King, G., Macy, M., Roy, D., & Van Alstyne, M. (2009). Life in the network: the coming age of computational social science. *Science*, 323(5915), 721-723. doi:10.1126/science.1167742.
- [6] Gilardi, F., Alizadeh, M., & Kubli, M. (2023). ChatGPT Outperforms Crowd-Workers for Text-Annotation Tasks. *arXiv:2303.15056*. doi:10.48550/arXiv.2303.15056.
- [7] Tan, Z., Li, D., Wang, S., Beigi, A., Jiang, B., Bhattacharjee, A., Karami, M., Li, J., Cheng, L., & Liu, H. (2024). Large Language Models for Data Annotation and Synthesis: A Survey. *Proceedings of EMNLP 2024*, 930-957. doi:10.18653/v1/2024.emnlp-main.54.

Appendices

Appendix A. Accompanying Files

The analysis scripts and supporting configuration files are provided as supplementary material rather than reproduced in full in the thesis body. The supplementary folder contains the project scripts.

project_scripts/

project_config/

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